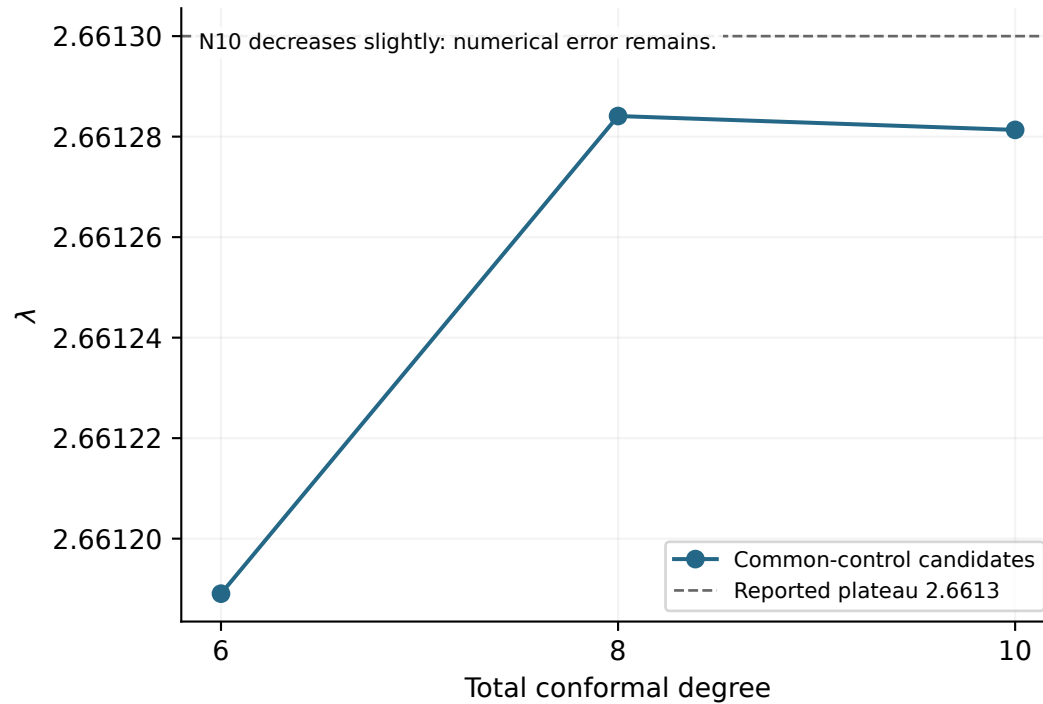


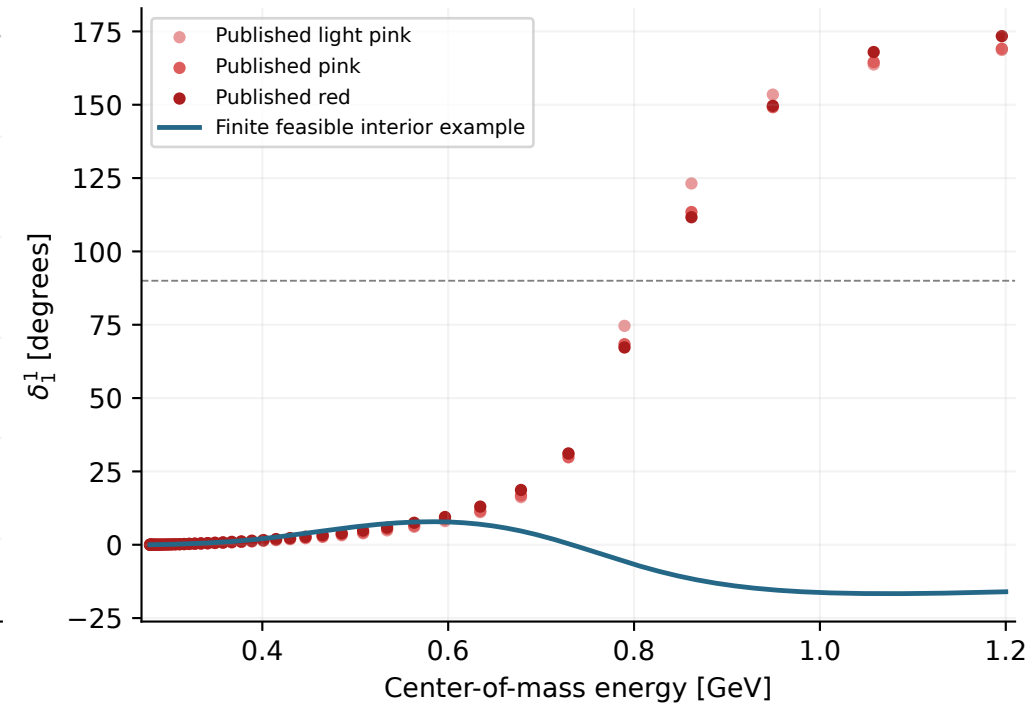
S-matrix bootstrap reproduction: verified progress and open physics

Scalar agreement is numerical; the coupled phase-shift goal remains active.

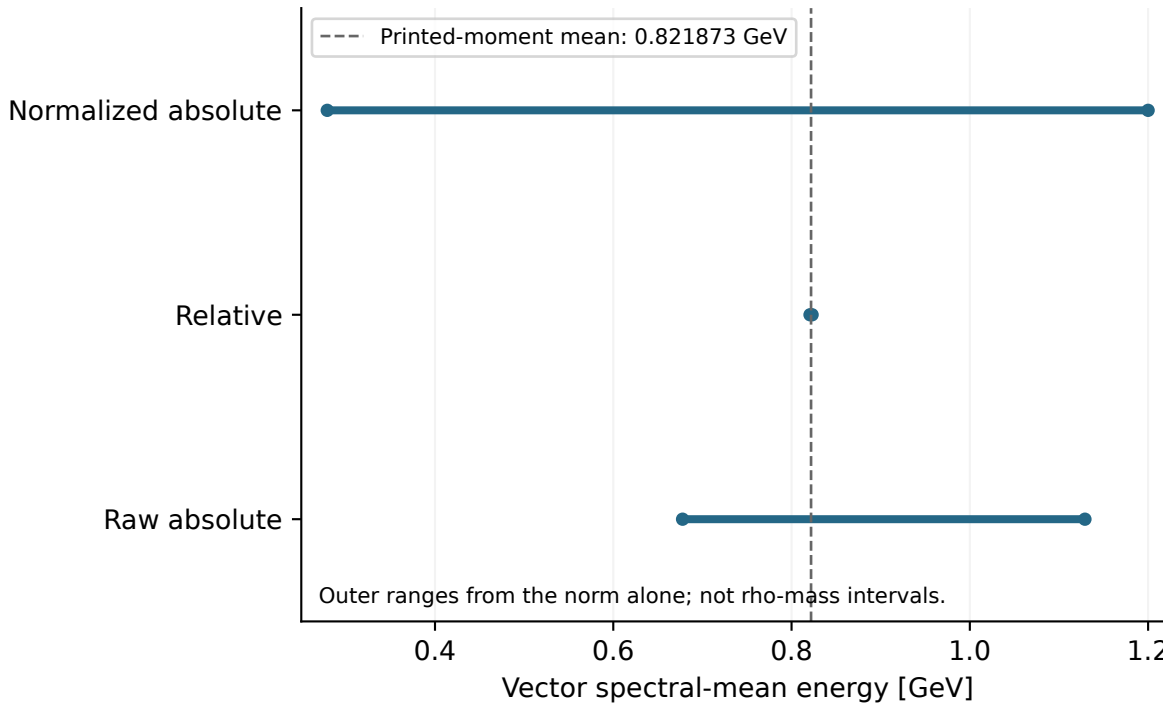
A. Scalar quartic benchmark



B. Gauge P1 phase remains unreproduced



C. Information retained by the moment norm



D. Exact statement for the gauge example

$$R_l(x) > 4.99 \times 10^{-4}$$

for all $x \in [-1, 1]$ and $l = 0, 1, 2$

Verified by rational Gram shifts and exact polynomial-reconstruction error bounds.

This certifies leading large-spin conditions.
It does not certify full unitarity.
Independent form-factor checks still fail.